Day 052 | Unit 04 | Teacher plan

Numeric and Algebraic Expressions

Identify Parts of Expressions

Standards: 6.EE.A.2b

Learning target: I can name terms, coefficients, factors, and grouped parts.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Addition and subtraction separate terms. A coefficient multiplies a variable; a constant has no variable. A parenthesized sum can be one factor of a larger product, so describe both the inside and the whole expression.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 14

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01 / Learn

Addition and subtraction separate terms. A coefficient multiplies a variable; a constant has no variable. A parenthesized sum can be one factor of a larger product, so describe both the inside and the whole expression.

Worked example 1

For 4x + 7y + 3, list terms, coefficients, and the constant.

1. Addition and subtraction separate terms.

2. A coefficient multiplies a variable; a constant has no variable.

3. Terms are separated by addition

4. coefficients multiply variables.

5. State the result in the form requested: Terms 4x, 7y, 3; coefficients 4, 7; constant 3.

Answer: Terms 4x, 7y, 3; coefficients 4, 7; constant 3

Worked example 2

For the expression 6(x + 2) + 4, select all true statements.
A. 6 and (x + 2) are factors of the first term.
B. 2 is the coefficient of x in the fully expanded expression.
C. 4 is a constant term.
D. The entire expression is a product with 4.

1. Addition and subtraction separate terms.

2. The outer sum has two terms.

3. The first term is a product with two factors.

4. Expanding gives 6x + 12 + 4, so the coefficient is 6.

Answer: A, C - 6 and (x + 2) are factors of the first term.; 4 is a constant term.

Day 052 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

For 4x + 4y + 6, list terms, coefficients, and the constant.

02 / Constructed response

For 3x + 7y + 5, list terms, coefficients, and the constant.

03 / Constructed response

For (6x + 7) ÷ 3, name the outer operation, the dividend as a single entity, and the divisor. Identify the sum inside the parentheses.

04 / Constructed response

For 2x + 5y + 2, list terms, coefficients, and the constant.

05 / Constructed response

For (4x + 4) ÷ 4, name the outer operation, the dividend as a single entity, and the divisor. Identify the sum inside the parentheses.

06 / Constructed response

For 6x + 5y + 5, list terms, coefficients, and the constant.

07 / Constructed response

For (2x + 3) ÷ 5, name the outer operation, the dividend as a single entity, and the divisor. Identify the sum inside the parentheses.

08 / Multiple select

For the expression 7(x + 4) + 2, select all true statements.
A. The entire expression is a product with 2.
B. 2 is a constant term.
C. 4 is the coefficient of x in the fully expanded expression.
D. 7 and (x + 4) are factors of the first term.

09 / Multiple select

For the expression 3(x + 4) + 3, select all true statements.
A. 4 is the coefficient of x in the fully expanded expression.
B. 3 is a constant term.
C. 3 and (x + 4) are factors of the first term.
D. The entire expression is a product with 3.

10 / Constructed response

In 7(x + 4), identify the outer operation, both factors, and the operation inside the grouped factor.

Day 052 | Unit 04 | Answer key

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1. Terms 4x, 4y, 6; coefficients 4, 4; constant 6

Terms are separated by addition; coefficients multiply variables.

6.EE.A.2b | Constructed response

2. Terms 3x, 7y, 5; coefficients 3, 7; constant 5

Terms are separated by addition; coefficients multiply variables.

6.EE.A.2b | Constructed response

3. Outer quotient; dividend (6x + 7); divisor 3; inner sum 6x + 7

The grouped sum is one complete quantity being divided. Within it, the terms are 6x and 7.

6.EE.A.2b | Constructed response

4. Terms 2x, 5y, 2; coefficients 2, 5; constant 2

Terms are separated by addition; coefficients multiply variables.

6.EE.A.2b | Constructed response

5. Outer quotient; dividend (4x + 4); divisor 4; inner sum 4x + 4

The grouped sum is one complete quantity being divided. Within it, the terms are 4x and 4.

6.EE.A.2b | Constructed response

6. Terms 6x, 5y, 5; coefficients 6, 5; constant 5

Terms are separated by addition; coefficients multiply variables.

6.EE.A.2b | Constructed response

7. Outer quotient; dividend (2x + 3); divisor 5; inner sum 2x + 3

The grouped sum is one complete quantity being divided. Within it, the terms are 2x and 3.

6.EE.A.2b | Constructed response

8. B, D - 2 is a constant term.; 7 and (x + 4) are factors of the first term.

The outer sum has two terms. The first term is a product with two factors. Expanding gives 7x + 28 + 2, so the coefficient is 7.

6.EE.A.2b | Multiple select

9. B, C - 3 is a constant term.; 3 and (x + 4) are factors of the first term.

The outer sum has two terms. The first term is a product with two factors. Expanding gives 3x + 12 + 3, so the coefficient is 3.

6.EE.A.2b | Multiple select

10. Outer product; factors 7 and (x + 4); inner sum

The complete sum (x + 4) is one factor multiplied by 7.

6.EE.A.2b | Constructed response